REMARKS ON THREE CONJECTURES ON POINT EVALUATION IN PALEY–WIENER SPACES

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ABSTRACT. Let $\mathscr{C}_p$ be the norm of point evaluation at the origin on the Paley–Wiener space PW^p, in the sense that $|f(0)|^p \le \mathscr{C}_p \|f\|_p^p$, and let φ_p be the extremal function. We address three conjectures of Brevig, Chirre, Ortega-Cerdà and Seip. First, by a computer-assisted proof, $\mathscr{C}_p \ge (p/2)(1-A(p-2))$ for $1 \le p \le 2$ with $A = 7/100$. Consequently $\mathscr{C}_p > p/2$ for every $0 < p < 2$, so the sign of $\mathscr{C}_p/p - 1/2$ is known on the whole half-line, and Korevaar's conjecture fails at every $1 \le p < 2$. Second, $\varphi_p(z) \to \cos \pi z$ locally uniformly as $p \to \infty$, so the nth positive zero of φ_p tends to $n-1/2$. Third, the zero set of φ_p together with the origin is a uniqueness set for PW^p when $p > 1$; for $p = 1$ this is contained in recent work of Bondarenko, Ortega-Cerdà, Radchenko and Seip, and we give a second proof via the asymptotics $\varphi_1(x) = -\cos \pi x/(2\pi \mathscr{C}_1 x^2) + O(|x|^{-3})$.

1. INTRODUCTION

For $0 < p < \infty$ let PW^p denote the Paley–Wiener space of entire functions of exponential type at most π whose restrictions to $\mathbb{R}$ belong to $L^p(\mathbb{R})$, equipped with

$$\|f\|_p^p = \int_{-\infty}^{\infty} |f(x)|^p \, \mathrm{d}x.$$

Let PW^∞ be the space of entire functions of exponential type at most π which are bounded on $\mathbb{R}$. Following Brevig, Chirre, Ortega-Cerdà and Seip [4], whose results we cite with the numbering of arXiv:2210.13922v4, we let $\mathscr{C}_p$ denote the smallest constant such that

$$|f(0)|^p \le \mathscr{C}_p \|f\|_p^p \tag{1}$$

for every f in PW^p. Equivalently,

$$\frac{1}{\mathscr{C}_p} = \inf\{\|f\|_p^p : \ f \in PW^p, \ f(0) = 1\}. \tag{2}$$

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For $1 \le p < \infty$ the extremal problem (2) has a unique solution φ_p, which is even and of the form

$$\varphi_p(z) = \prod_{n=1}^{\infty} \Big(1 - \frac{z^2}{t_n^2}\Big),$$

where $0 < t_1 < t_2 < \cdots$ [4, Lemma 3.5]. We write $t_n = t_n(p)$ and $\mathscr{Z}(\varphi_p) = \{\pm t_n : \ n \ge 1\}$.

The reproducing formula for PW^2 and the Cauchy–Schwarz inequality give $\mathscr{C}_2 = 1$, with $\varphi_2(z) = \operatorname{sinc} \pi z$, where $\operatorname{sinc} z = (\sin z)/z$. Hörmander and Bernhardsson [5] computed $\mathscr{C}_1$ to ten decimal places; in particular $\mathscr{C}_1 > 0.5409$. Brevig, Chirre, Ortega-Cerdà and Seip proved that $\mathscr{C}_p < p/2$ for every $p > 2$, found the asymptotic behaviour of $\mathscr{C}_p$ as $p \to \infty$ and as $p \to 0^+$, and studied the zeros of φ_p in detail [4]. Their *monotonicity conjecture* [4, Conjecture 9.1] asserts that $p \mapsto \mathscr{C}_p/p$ is strictly decreasing on $(0, \infty)$. In the range $1 \le p < 2$ they verified numerically, but did not prove, that $\mathscr{C}_p > p/2$, and they made the following quantitative conjecture.

Conjecture ([4, Conjecture 9.2])**.** *There is a positive constant A such that $\mathscr{C}_p \ge (p/2)(1 - A(p-2))$ for $1 \le p \le 2$.*

The main purpose of the present paper is to prove this conjecture with an explicit constant. The proof is computer-assisted. The closest prior work is that of Brevig, Chirre, Ortega-Cerdà and Seip themselves [4], who verified $\mathscr{C}_p > p/2$ numerically for $1 \le p < 2$; in that range it was previously proved only at $p = 1$, where it follows from the bound $\mathscr{C}_1 > 0.5409$ of Hörmander and Bernhardsson [5]. On the other side of $p = 2$, Instanes [6] improved the bound $\mathscr{C}_p < p/2$ for $2 < p \le 5$ by solving an optimization problem.

Theorem 1 (computer-assisted)**.** *Suppose that $1 \le p \le 2$. Then*

$$\mathscr{C}_p \ge \frac{p}{2}\Big(1 - \frac{7}{100}(p-2)\Big).$$

The only part of the proof which relies on a computer is the evaluation, in ball arithmetic, of upper bounds for fifteen integrals and of finitely many elementary inequalities; see Section 4. Combining Theorem 1 with the power trick of [4, Lemma 2.1], we settle the sign of $\mathscr{C}_p/p - 1/2$ on the whole half-line.

Corollary 1. *Suppose that $0 < p < 2$. Then $\mathscr{C}_p > p/2$. If moreover $p < 1$, then $\mathscr{C}_p \ge (p/2)(1 + (7/100)(1-p))$.*

Together with $\mathscr{C}_2 = 1$ and $\mathscr{C}_p < p/2$ for $p > 2$ [4, Theorem 1.1 and Lemma 2.1], Corollary 1 shows that $\mathscr{C}_p/p - 1/2$ is positive on $(0, 2)$, vanishes at $p = 2$ and is negative on $(2, \infty)$, as predicted by the monotonicity conjecture. Corollary 1 also bears on a problem of Korevaar [9]. Let $\mathscr{K}_p$ be the smallest constant K such that

$$|f(iy)|^p \le K \operatorname{sinc}(i\pi p y) \|f\|_p^p$$

for every $y \ge 0$ and every f in PW^p. Taking $y = 0$ we see that $\mathscr{C}_p \le \mathscr{K}_p$. Korevaar conjectured that $\mathscr{K}_p = p/2$ for $1 \le p < \infty$, and Brevig, Chirre,

Ortega-Cerdà and Seip observed that this fails at $p = 1$ in view of the bound $\mathscr{C}_1 > 0.5409$ [4, Section 9.2]. By Corollary 1, $\mathscr{K}_p \geq \mathscr{C}_p > p/2$ for every $0 < p < 2$. In particular Korevaar's conjecture fails at every $1 \leq p < 2$. This is consistent with [4, Conjecture 9.3], which predicts that $\mathscr{K}_p = \mathscr{C}_p$ for $0 < p < 2$.

Our second result concerns the zeros of φ_p as $p \to \infty$. Brevig, Chirre, Ortega-Cerdà and Seip conjectured that $t_1(p) \to 1/2$ as $p \to \infty$ and that $\lim_{p\to 0^+} p\, t_1(p) > 0$ [4, Conjecture 9.10]. We prove the first half in the following stronger form.

Theorem 2. *Let $1 \leq p < \infty$, and let φ_p and $t_n(p)$ be as above. As $p \to \infty$, the following hold.*

(a) $0 \leq \pi^2 + \varphi_p''(0) = O\big((\log p)/p\big)$.
(b) $\varphi_p(z) \to \cos \pi z$ *locally uniformly in* $\mathbb{C}$.
(c) $t_n(p) \to n - 1/2$ *for every* $n \geq 1$. *In particular,* $\lim_{p\to\infty} t_1(p) = 1/2$.

The second half of [4, Conjecture 9.10], which concerns $p \to 0^+$, is not addressed here. Our third result concerns uniqueness sets. A set $\Lambda \subset \mathbb{C}$ is a *uniqueness set* for PW^p if the only function in PW^p which vanishes on Λ is the zero function. Since $x\varphi_2(x) = \pi^{-1} \sin \pi x$, the set $\mathscr{Z}(\varphi_p) \cup \{0\}$ is a natural analogue of the integers.

Conjecture ([4, Conjecture 9.11])**.** *The set $\mathscr{Z}(\varphi_p) \cup \{0\}$ is a uniqueness set for PW^p if and only if $p \geq 1$.*

Brevig, Chirre, Ortega-Cerdà and Seip noted that the case $p \geq 2$ follows from [4, Lemma 8.4], and singled out $1 < p < 2$ as the most accessible open range.

Theorem 3. *Suppose that $1 \leq p < \infty$. Then $\mathscr{Z}(\varphi_p) \cup \{0\}$ is a uniqueness set for PW^p.*

Hence the "if" direction of [4, Conjecture 9.11] holds. The "only if" direction, for $p < 1$, is not addressed here. The case $p = 1$ is contained in the work of Bondarenko, Ortega-Cerdà, Radchenko and Seip on the Hörmander–Bernhardsson function: by [3, Lemma 2.5], which applies by [2, Theorem 1], every $f \in PW^1$ vanishing on $\mathscr{Z}(\varphi_1)$ is a multiple of φ_1, and $\varphi_1(0) = 1$. Our contribution is the case $p > 1$; for $p = 1$ we include a second proof through their functional equation (see Section 7).

Let us briefly describe our approach. For Theorem 1 we test (1) with the functions

$$g_\alpha(z) = \frac{\Gamma^2(\alpha)}{\Gamma(\alpha - z)\Gamma(\alpha + z)}$$

of [4, Section 7.1], whose zeros are known explicitly. By Hölder's inequality, $p \mapsto \log \|g_\alpha\|_p^p$ is convex, while the logarithm of the target bound is concave in p. Hence finitely many upper bounds for $\|g_\alpha\|_p^p$, at the endpoints of finitely many intervals in p, suffice for the whole interval $1 \leq p \leq 2$. Near $p = 2$ we use $g_1(z) = \operatorname{sinc} \pi z$ and the exact value $\|g_1\|_2 = 1$. For Theorem 2, we show

that $\varphi_p''(0) \to -\pi^2$ by comparing a Laplace type lower bound for $\|\varphi_p\|_p^p$ with the asymptotic lower bound for $\mathscr{C}_p$ from [4, Theorem 1.2]. A second-order version of the Riesz interpolation formula then forces every limit function to be $\cos \pi z$, and Hurwitz's theorem locates the zeros. For Theorem 3, a function in PW^p vanishing on $\mathscr{Z}(\varphi_p) \cup \{0\}$ has the form $\omega \cdot x\varphi_p$ with ω entire. The lower bound for $|\varphi_p|$ from [4, (8.4)] and a Poisson majorant show that ω is a polynomial, and (B3) shows that it is constant. When $p > 1$, the reproducing formula [4, Theorem 3.6], applied to the derivative of $x\varphi_p$, shows that $x\varphi_p$ is not in $L^p(\mathbb{R})$, which excludes the nonzero constants. When $p = 1$, we obtain the same conclusion from the asymptotic formula $\varphi_1(x) = -\cos \pi x/(2\pi \mathscr{C}_1 x^2) + O(|x|^{-3})$, which we derive from the functional equation of [3, Theorem 1.1].

Organization. The present paper is organized into six additional sections. In Section 2 we collect the facts from [4] and the classical results that we use. In Section 3 we reduce Theorem 1 to finitely many inequalities, and in Section 4 we describe how these are verified in ball arithmetic and prove Theorem 1 and Corollary 1. Section 5 contains the proof of Theorem 2, and Section 6 the proof of Theorem 3, including the asymptotic formula for φ_1 in Lemma 8. Section 7 contains some concluding remarks, among them an evaluation of the integral which limits the constant A attainable by our method.

2. Preliminaries

Functions in PW^∞. We use three classical facts. Suppose that f is in PW^∞. Then

$$|f(x+iy)| \le \|f\|_\infty e^{\pi|y|} \tag{3}$$

by the Phragmén–Lindelöf principle, and Bernstein's inequality $\|f'\|_\infty \le \pi\|f\|_\infty$ holds [10, Lecture 28]. In particular f' is in PW^∞, and $\|f^{(k)}\|_\infty \le \pi^k\|f\|_\infty$ for every $k \ge 0$. Finally, the Riesz interpolation formula [1, Section 11.3] states that

$$f'(x_0) = \frac{1}{\pi} \sum_{n=-\infty}^{\infty} a_n f(x_0 + n + 1/2), \qquad a_n = \frac{(-1)^n}{(n+1/2)^2}, \tag{4}$$

for every real x_0. This is [4, (7.6)]. Note that $\sum_n |a_n| = \pi^2$.

The extremal functions. Let $1 \le p < \infty$. Applying (1) to the translates $\varphi_p(\cdot + x)$ and using $\|\varphi_p\|_p^p = 1/\mathscr{C}_p$, we find that $|\varphi_p(x)| \le 1 = \varphi_p(0)$ for every real x. Hence φ_p is in PW^∞ with $\|\varphi_p\|_\infty = 1$. We will use the following results from [4].

(B1) The zeros $\pm t_n$ are real and simple, $\mathscr{Z}(\varphi_p)$ has a positive separation constant σ, and $t_n/n \to 1$ as $n \to \infty$ [4, Theorem 1.4 and Lemma 3.5, and the proof of Lemma 8.3].

(B2) There is $C > 0$ such that $|\varphi_p(x)| \geq C \operatorname{dist}(x, \mathscr{Z}(\varphi_p))(1+|x|)^{-4/p}$ for every real x [4, (8.4)].
(B3) $\int_{\mathbb{R}} x^2 |\varphi_p(x)|^p \,\mathrm{d}x = \infty$ [4, Corollary 3.7].
(B4) If $p > 1$ and f is in PW^p, then [4, Theorem 3.6]

$$f(0)\|\varphi_p\|_p^p = \int_{\mathbb{R}} f(x)|\varphi_p(x)|^{p-2}\varphi_p(x)\,\mathrm{d}x.$$

(B5) There is a constant A_0 such that $\mathscr{C}_p \geq \sqrt{\pi p/2} - A_0(\log p)/\sqrt{p}$ for all sufficiently large p [4, Theorem 1.2].
(B6) If $0 < p < \infty$ and k is a positive integer, then $\mathscr{C}_{kp}/(kp) \leq \mathscr{C}_p/p$ [4, Lemma 2.1].

Test functions. For $\alpha > 1/2$, Brevig, Chirre, Ortega-Cerdà and Seip [4, Section 7.1] consider

$$g_\alpha(z) = \frac{\Gamma^2(\alpha)}{\Gamma(\alpha - z)\Gamma(\alpha + z)} = \prod_{n=1}^{\infty}\Big(1 - \frac{z^2}{(n+\alpha-1)^2}\Big).$$

This is a real even entire function of exponential type π with $g_\alpha(0) = 1$, it is positive on $(-\alpha, \alpha)$, and its zeros are $\pm(n+\alpha-1)$, $n \geq 1$. Moreover $g_1(z) = \operatorname{sinc} \pi z$. By the reflection formula for the gamma function [4, (7.3)],

$$g_\alpha(x) = \frac{\Gamma^2(\alpha)}{\pi}\,\frac{\Gamma(1-\alpha+x)}{\Gamma(\alpha+x)}\sin\pi(\alpha - x) \tag{5}$$

for $x > \alpha - 1$.

3. Reduction of Theorem 1 to finitely many inequalities

Throughout this section and the next, fix $A = 7/100$ and set

$$H(p) = \log\frac{2}{p(1 + A(2-p))}.$$

Note that $1 - A(p-2) = 1 + A(2-p)$. For $1 \leq q \leq 2$, put $\alpha = 1/2 + 1/q$ and $s = 2\alpha - 1 = 2/q$, and define

$$J_q(p) = \int_{-\infty}^{\infty} |g_\alpha(x)|^p \,\mathrm{d}x.$$

Since $g_\alpha(0) = 1$, (1) gives $\mathscr{C}_p \geq 1/J_q(p)$ whenever $J_q(p) < \infty$. Hence the bound of Theorem 1 holds at p as soon as $\log J_q(p) \leq H(p)$ for some q.

The first lemma records the formula for g_α that is used in the computations, and a bound for the tail of $J_q(p)$. We split $[\alpha, \infty)$ into the cells $[\alpha+n-1, \alpha+n]$, $n \geq 1$, between consecutive zeros, and write $m_p = \int_0^1 \sin^p(\pi t)\,\mathrm{d}t$.

Lemma 1. *Suppose that $1 \leq q \leq 2$, and let $K = \Gamma^2(\alpha)/\pi$. Then*

$$|g_\alpha(\alpha + n - 1 + t)| = K\,\frac{\Gamma(n+t)}{\Gamma(n+t+s)}\sin\pi t$$

for every integer $n \geq 1$ and every $0 \leq t \leq 1$. Suppose moreover that $sp > 1$ and that $N \geq 1$ is an integer. Then

$$\sum_{n=N}^{\infty} \int_0^1 |g_\alpha(\alpha + n - 1 + t)|^p \, \mathrm{d}t \leq K^p m_p \Big(N^{-sp} + \frac{N^{1-sp}}{sp-1} \Big). \tag{6}$$

In particular $J_q(p) < \infty$, and g_α is in PW^p.

Proof. The formula follows from (5) with $x = \alpha + n - 1 + t$, since $1 - \alpha + x = n + t$, $\alpha + x = n + t + s$ and $|\sin \pi(1 - n - t)| = \sin \pi t$. Let $R(y) = \Gamma(y)/\Gamma(y+s)$ for $y > 0$. Since the digamma function is increasing, $(\log R)'(y) = \psi(y) - \psi(y+s) < 0$, so R is decreasing. We next show that $R(n) \leq n^{-s}$. Write $s = 1 + s'$ with $0 \leq s' \leq 1$. Since $n + 1 = s'(n + s') + (1 - s')(n + s' + 1)$, the logarithmic convexity of Γ gives

$$n\Gamma(n) = \Gamma(n+1) \leq \Gamma(n+s')^{s'} \Gamma(n+s'+1)^{1-s'} = \Gamma(n+s')(n+s')^{1-s'}.$$

Hence

$$R(n) = \frac{\Gamma(n)}{(n+s')\Gamma(n+s')} \leq \frac{1}{n(n+s')^{s'}} \leq n^{-s}.$$

Since R is decreasing, the term with index n on the left-hand side of (6) is at most $K^p R(n)^p m_p$, which is at most $K^p m_p n^{-sp}$, and (6) follows by comparing the sum of n^{-sp} over $n > N$ with an integral. Since g_α is even and bounded, $J_q(p) < \infty$. □

The next lemma is the convexity argument which reduces the continuum of exponents in a patch $a \leq p \leq b$ to the two endpoints.

Lemma 2. *Suppose that $1 \leq a < b \leq 2$, that $1 \leq q \leq 2$ with $2a > q$, and that $U_a \geq J_q(a)$ and $U_b \geq J_q(b)$. Let ℓ be the affine function with $\ell(a) = \log U_a$ and $\ell(b) = \log U_b$, and set $\Delta = H - \ell$. Then Δ is convex on $[a, b]$. If $\Delta \geq 0$ on $[a, b]$, then $\mathscr{C}_p \geq (p/2)(1 + A(2-p))$ for every $a \leq p \leq b$.*

Proof. Since $sa = 2a/q > 1$, Lemma 1 shows that $J_q(p)$ is finite for $a \leq p \leq b$. If $p = (1-\theta)a + \theta b$ with $0 \leq \theta \leq 1$, then Hölder's inequality with exponents $1/(1-\theta)$ and $1/\theta$ gives $J_q(p) \leq J_q(a)^{1-\theta} J_q(b)^\theta$. Hence $\log J_q(p) \leq (1-\theta)\log U_a + \theta \log U_b = \ell(p)$. If $\Delta \geq 0$ on $[a, b]$, then $\log J_q(p) \leq H(p)$ there, which is the assertion. Finally,

$$H''(p) = \frac{1}{p^2} + \frac{A^2}{\big(1 + A(2-p)\big)^2} > 0,$$

and ℓ is affine, so Δ is convex. □

To check that $\Delta \geq 0$ on $[a, b]$ we use convexity once more. Suppose that $a \leq c < d \leq b$. Since $\Delta(p) \geq \Delta(c) + \Delta'(c)(p - c)$ and $0 \leq p - c \leq d - c$, we have

$$\Delta(p) \geq \Delta(c) + \min\big(0, \Delta'(c)\big)(d - c) \tag{7}$$

for $c \leq p \leq d$. Hence $\Delta \geq 0$ on $[a, b]$ follows from finitely many evaluations of Δ and Δ' on a subdivision of $[a, b]$.

The patch which contains $p = 2$ is treated differently, because the bound of Theorem 1 is attained at $p = 2$. Here we use $g_1(z) = \operatorname{sinc} \pi z$, which corresponds to $q = 2$, together with the identity $J_2(2) = \|\operatorname{sinc} \pi x\|_2^2 = 1$ from Plancherel's theorem.

Lemma 3. *Suppose that $1 < a < 2$, that $U \geq J_2(a)$, and that*

$$-\frac{1}{2} + A + \frac{\log U}{2-a} \leq 0. \tag{8}$$

Then $\mathscr{C}_p \geq (p/2)\big(1 + A(2-p)\big)$ for every $a \leq p \leq 2$.

Proof. Let ℓ be the affine function with $\ell(a) = \log U$ and $\ell(2) = 0$, and set $\Delta = H - \ell$. As in the proof of Lemma 2, with $U_b = J_2(2) = 1$, we have $\log J_2(p) \leq \ell(p)$ for $a \leq p \leq 2$, and Δ is convex. Moreover $\Delta(2) = H(2) - 0 = 0$ and

$$\Delta'(2) = H'(2) + \frac{\log U}{2-a} = -\frac{1}{2} + A + \frac{\log U}{2-a} \leq 0.$$

Since Δ' is increasing, $\Delta' \leq 0$ on $[a, 2]$. Hence Δ is decreasing there, and $\Delta(p) \geq \Delta(2) = 0$ for $a \leq p \leq 2$. As in the proof of Lemma 2, this gives the assertion. □

4. The certificate and the proof of Theorem 1

Lemmas 2 and 3 reduce Theorem 1 to a cover of $[1, 2]$ by patches. For $A = 7/100$ we use seven patches plus the sinc patch: the seven patches in Table 1 with the functions $g_{1/2+1/q}$, followed by the patch $[1.977, 2]$ with $g_1 = \operatorname{sinc} \pi z$. In every patch $2a > q$, with a wide margin.

What is computed. All quantities below were computed in ball (midpoint-radius interval) arithmetic with the Arb library [7] through python-flint, version 0.9.0, at a working precision of 80 bits. The parameters p, q and A are exact rational numbers and enter the computation as balls which contain them. Every ball returned by Arb is guaranteed to contain the exact value of the quantity it represents, so every inequality between balls below is a proof of the corresponding inequality between exact values.

(i) *Upper bounds for $J_q(p)$.* Since g_α is even,

$$J_q(p) = 2\int_0^\alpha g_\alpha(x)^p \,\mathrm{d}x + 2\sum_{n=1}^{N-1} \int_0^1 |g_\alpha(\alpha + n - 1 + t)|^p \,\mathrm{d}t + 2\sum_{n=N}^{\infty} \int_0^1 |g_\alpha(\alpha + n - 1 + t)|^p \,\mathrm{d}t.$$

We take $N = 400$, and $N = 2000$ in the sinc patch. The last sum is bounded by (6), where K^p, $m_p = \Gamma((p+1)/2)/(\sqrt{\pi}\,\Gamma(p/2+1))$ and the powers of N are evaluated in ball arithmetic. The first N

integrals are enclosed by Arb's rigorous integration routine [8], with relative tolerance 2^{-40}. The integrand on the cell n is the function

$$t \mapsto \exp\bigl(p\bigl(\log K + \log\Gamma(n+t) - \log\Gamma(n+t+s) + \log\sin\pi t\bigr)\bigr),$$

with principal branches, which equals $|g_\alpha(\alpha+n-1+t)|^p$ for $0 < t < 1$. On $[0,\alpha]$ we substitute $x = \alpha u$ and integrate the analogous expression for $\alpha g_\alpha(\alpha u)^p$ over $0 \le u \le 1$, with $1/\Gamma(\alpha \pm \alpha u)$ in place of the quotient of gamma functions. The integration routine either applies Gauss–Legendre quadrature with an error bound, which requires the integrand to be holomorphic on a neighbourhood of the subinterval (Arb certifies this by checking that no ball meets a branch cut), or it bounds the integral by the length of the subinterval times an enclosure of the integrand. Near the zeros $t = 0$ and $t = 1$ we use the enclosure $0 \le |g_\alpha|^p \le |e^{p\log K}|\, R(n+t)^p |\sin\pi t|^p$. The same crude enclosure, with $1/\Gamma(\alpha - \alpha u)$ in place of $\sin\pi t$, covers the inner cell near its zero $u = 1$. Only the upper end of the resulting ball is used.

(ii) *The inequality* $\Delta \ge 0$. For each of the seven patches $[a,b]$ in Table 1, we split $[a,b]$ into 64 intervals $[c,d]$ of equal length, with rational endpoints, and evaluate the right-hand side of (7) with U_a and U_b from (i). The table lists the smallest lower bound obtained. If this bound were not positive on some $[c,d]$, the script would bisect $[c,d]$, up to depth 12; this fallback is not used in the reported runs.

(iii) *The condition* (8) with $a = 1.977$ and the upper bound U for $J_2(a)$ from (i).

The computation takes about a minute on one core, and every check in (ii) and (iii) succeeds. As a check of (i), the same routine encloses $J_2(2)$ in an interval of length less than $3 \cdot 10^{-4}$ which contains the exact value 1.

The rounded values in Table 1 suffice on their own: with these values of U_a and U_b, the evaluation of (7) on 64 equal subintervals and the evaluation of (8) are elementary, and they give margins which differ from the last column by less than $3 \cdot 10^{-6}$. The only nontrivial computation is therefore the fifteen upper bounds for $J_q(p)$ in (i).

Code availability. The certificate is a Python script which requires python-flint 0.9.0. It is available at `https://github.com/Oisov/math-preprints`, in the directory `numerics/paley-wiener/`. Run without arguments it proves Theorem 1 with $A = 7/100$; with the argument `0.06` it proves the case $A = 3/50$ discussed in Section 7.

Proof of Theorem 1. The patches in Table 1 cover $[1,2]$. On each of the first seven, the hypotheses of Lemma 2 hold by (i) and (ii). On the last one, the hypotheses of Lemma 3 hold by (i) and (iii). Hence $\mathscr{C}_p \ge (p/2)\bigl(1 + A(2-p)\bigr)$ for every $1 \le p \le 2$. □

Proof of Corollary 1. If $1 \le p < 2$, then $\mathscr{C}_p \ge (p/2)\bigl(1+(7/100)(2-p)\bigr) > p/2$ by Theorem 1. Suppose that $0 < p < 1$ and let $k = \lceil 1/p \rceil$. Then $k \ge 1/p$ and

TABLE 1. The patches for Theorem 1. The columns U_a and U_b are the certified upper bounds for $J_q(a)$ and $J_q(b)$, rounded up, and the last column is the certified lower bound for $\min_{[a,b]} \Delta$, rounded down. In the sinc patch, the last column is the certified upper bound for the left-hand side of (8), rounded up. The value $U_b = 1$ in the last row is the exact value $J_2(2) = 1$.

q	a	b	U_a	U_b	margin
1.122	1	1.182	1.8608319	1.5960498	$5.1 \cdot 10^{-5}$
1.314	1.182	1.37	1.5958174	1.3943834	$5.7 \cdot 10^{-5}$
1.471	1.37	1.548	1.3924036	1.2517219	$3.6 \cdot 10^{-5}$
1.655	1.548	1.713	1.2499918	1.1435817	$2.6 \cdot 10^{-5}$
1.801	1.713	1.847	1.1433170	1.0708705	$1.8 \cdot 10^{-5}$
1.912	1.847	1.941	1.0708369	1.0259491	$1.8 \cdot 10^{-5}$
1.953	1.941	1.977	1.0257176	1.0099772	$3.0 \cdot 10^{-5}$
2	1.977	2	1.0099347	1	$-1.9 \cdot 10^{-4}$

$k < 1/p + 1$, so $q = kp$ satisfies $1 \le q < 1 + p < 2$. By (B6) and Theorem 1,

$$\frac{\mathscr{C}_p}{p} \ge \frac{\mathscr{C}_q}{q} \ge \frac{1}{2}\Big(1 + \frac{7}{100}(2-q)\Big) > \frac{1}{2}\Big(1 + \frac{7}{100}(1-p)\Big),$$

since $2 - q > 1 - p$. In particular $\mathscr{C}_p > p/2$. □

5. The extremal functions as $p \to \infty$

In this section $1 \le p < \infty$, and we write $c_p = -\varphi_p''(0)$. Since φ_p is real and even with a maximum at the origin, and since $\|\varphi_p''\|_\infty \le \pi^2$ by Bernstein's inequality, we have $0 \le c_p \le \pi^2$.

The first step is a lower bound for $\|f\|_p^p$ in terms of $f''(0)$, which is uniform over the functions considered. It is a quantitative form of Laplace's method.

Lemma 4. *Suppose that f is a real and even function in PW^∞ with $\|f\|_\infty = f(0) = 1$, and let $\gamma = \max(-f''(0), 1)$. If $p \ge 38$, then*

$$\|f\|_p^p \ge \sqrt{\frac{2\pi}{p\gamma}}\Big(1 - \frac{162}{p} - e^{-p^{1/3}/2}\Big).$$

Proof. Set $\kappa = \pi^4/24$ and $c = -f''(0)$. By Bernstein's inequality, $\|f''\|_\infty \le \pi^2$ and $\|f^{(4)}\|_\infty \le \pi^4$. In particular $0 \le c \le \pi^2$, so $1 \le \gamma \le \pi^2$. Since f is even, $f'(0) = f^{(3)}(0) = 0$, and Taylor's formula gives

$$f(x) \ge 1 - \frac{c}{2}x^2 - \kappa x^4 \ge 1 - u(x), \qquad u(x) = \frac{\gamma}{2}x^2 + \kappa x^4.$$

Let $\delta = p^{-1/3}$, so that $\delta < 0.3$. For $|x| \le \delta$ we have $u(x) \le 0.09\pi^2/2 + 0.0081\kappa < 1/2$, and hence $f(x) > 0$. Since $\log(1-u) \ge -u - u^2$ for

$0 \le u \le 1/2$ and $e^{-v} \ge 1 - v$, we get

$$f(x)^p \ge e^{-p\gamma x^2/2} e^{-p(\kappa x^4 + u(x)^2)} \ge e^{-p\gamma x^2/2}(1 - p(\kappa x^4 + u(x)^2))$$

for $|x| \le \delta$. Moreover $u(x)^2 \le \gamma^2 x^4/2 + 2\kappa^2 x^8 \le (\pi^4/2 + 2\kappa^2\delta^4)x^4$, and $2\kappa^2\delta^4 < 1$. Hence $\kappa x^4 + u(x)^2 \le Lx^4$ with $L = \kappa + \pi^4/2 + 1 < 54$. Set $\beta = p\gamma$. Then

$$\|f\|_p^p \ge \int_{-\delta}^{\delta} f(x)^p \,\mathrm{d}x \ge \int_{-\infty}^{\infty} e^{-\beta x^2/2} \,\mathrm{d}x - \int_{|x|>\delta} e^{-\beta x^2/2} \,\mathrm{d}x - pL \int_{-\infty}^{\infty} x^4 e^{-\beta x^2/2} \,\mathrm{d}x.$$

The first integral is $\sqrt{2\pi/\beta}$. By the Gaussian tail bound $\operatorname{erfc}(y) \le e^{-y^2}$, the second integral is at most $\sqrt{2\pi/\beta}\, e^{-\beta\delta^2/2} \le \sqrt{2\pi/\beta}\, e^{-p^{1/3}/2}$, since $\beta\delta^2 = \gamma p^{1/3} \ge p^{1/3}$. The third integral equals $3\beta^{-2}\sqrt{2\pi/\beta}$, and $3pL\beta^{-2} = 3L/(p\gamma^2) \le 162/p$. □

The next lemma is a second-order version of (4), which we use to identify the limits of φ_p.

Lemma 5. *There are real numbers w_m, $m \in \mathbb{Z}$, such that $(-1)^{m-1} w_m > 0$ for every m, $\sum_m |w_m| = \pi^2$, and*

$$f''(0) = \sum_{m=-\infty}^{\infty} w_m f(m)$$

for every f in PW^∞.

Proof. Let f be in PW^∞. Then f' is in PW^∞, and two applications of (4) give

$$f''(0) = \frac{1}{\pi} \sum_j a_j f'(j + 1/2) = \frac{1}{\pi^2} \sum_j \sum_k a_j a_k f(j + k + 1).$$

The double series converges absolutely since $\sum_j |a_j| < \infty$ and f is bounded on $\mathbb{R}$. Grouping the terms according to $m = j + k + 1$, we obtain the assertion with $w_m = \pi^{-2} \sum_{j+k+1=m} a_j a_k$. Each term in this sum is nonzero and has the sign $(-1)^{j+k} = (-1)^{m-1}$, and $\sum_m |w_m| = \pi^{-2} \bigl(\sum_j |a_j|\bigr)^2 = \pi^2$. □

Lemma 6. *Suppose that Φ is a function in PW^∞ which is real on $\mathbb{R}$, and that $\|\Phi\|_\infty \le 1$ and $\Phi''(0) = -\pi^2$. Then $\Phi(z) = \cos \pi z$.*

Proof. By Lemma 5,

$$-\pi^2 = \Phi''(0) = \sum_m w_m \Phi(m) \ge -\sum_m |w_m| = -\pi^2,$$

so $w_m \Phi(m) = -|w_m|$ for every m. Since $w_m \ne 0$, we find that $\Phi(m) = -\operatorname{sgn} w_m = (-1)^m$. Hence $h(z) = \Phi(z) - \cos \pi z$ is in PW^∞ and vanishes on the integers, and $G(z) = h(z)/\sin \pi z$ is entire. Let $N \ge 1$ be an integer and let Q_N be the square with vertices $\pm(N + 1/2) \pm i(N + 1/2)$. On the vertical sides of Q_N we have $|\sin \pi z| = \cosh \pi y$, and on the horizontal sides

$|\sin \pi z| \geq \sinh \pi |y|$ with $|y| \geq 3/2$. Hence $|\sin \pi z| \geq (1-e^{-\pi})e^{\pi|y|}/2$ on ∂Q_N, while $|h(z)| \leq 2e^{\pi|y|}$ by (3). By the maximum principle, $|G| \leq 4/(1-e^{-\pi})$ on Q_N for every N, so G is a constant b by Liouville's theorem, and b is real since h is real on $\mathbb{R}$. Hence $\Phi(x) = \cos \pi x + b \sin \pi x$, and $\sup_{x\in\mathbb{R}} |\Phi(x)| = \sqrt{1+b^2} \leq 1$ gives $b = 0$. □

We are now ready to prove Theorem 2.

Proof of Theorem 2. (a) Let $\gamma_p = \max(c_p, 1)$ and $\varepsilon_p = 162/p + e^{-p^{1/3}/2}$. Lemma 4 applies to φ_p, and gives $1/\mathscr{C}_p = \|\varphi_p\|_p^p \geq \sqrt{2\pi/(p\gamma_p)}\,(1-\varepsilon_p)$ for $p \geq 38$. By (B5), $\mathscr{C}_p \geq \sqrt{\pi p/2}\,(1-\eta_p)$ for large p, where $\eta_p = A_0\sqrt{2/\pi}\,(\log p)/p$. Combining these two bounds, we find that

$$\frac{\pi}{\sqrt{\gamma_p}} \leq \frac{1}{(1-\varepsilon_p)(1-\eta_p)},$$

and hence $\gamma_p \geq \pi^2(1-\varepsilon_p)^2(1-\eta_p)^2 \geq \pi^2(1-2\varepsilon_p-2\eta_p)$. For large p the right-hand side exceeds 1, so $\gamma_p = c_p$. Since $c_p \leq \pi^2$, we obtain $0 \leq \pi^2 - c_p \leq 2\pi^2(\varepsilon_p+\eta_p) = O((\log p)/p)$.

(b) By (3), $|\varphi_p(x+iy)| \leq e^{\pi|y|}$ for every p, so the family $\{\varphi_p : p \geq 1\}$ is locally bounded in $\mathbb{C}$. Let $p_k \to \infty$. By Montel's theorem, a subsequence, which we do not relabel, converges locally uniformly in $\mathbb{C}$ to an entire function Φ, and the derivatives converge as well. Then $|\Phi(x+iy)| \leq e^{\pi|y|}$, so Φ is in PW^∞, and Φ is real on $\mathbb{R}$ with $\|\Phi\|_\infty \leq 1$. By (a), $\Phi''(0) = \lim_k(-c_{p_k}) = -\pi^2$, and Lemma 6 gives $\Phi(z) = \cos \pi z$. Since every sequence $p_k \to \infty$ has a subsequence along which $\varphi_{p_k} \to \cos \pi z$, we conclude that $\varphi_p \to \cos \pi z$ locally uniformly as $p \to \infty$.

(c) Fix $n \geq 1$ and $0 < r < 1/4$. The zeros of $\cos \pi z$ in the closed disc $D = \{|z| \leq n\}$ are the simple zeros $\pm(k-1/2)$, $1 \leq k \leq n$, and $\cos \pi z \neq 0$ on ∂D. Let $D_k^\pm$ be the open disc of radius r centred at $\pm(k-1/2)$. By (b) and Hurwitz's theorem, for all sufficiently large p the function φ_p has exactly one zero in each $D_k^\pm$ and no zeros in the compact set $D \setminus \bigcup_k D_k^\pm$. The zeros of φ_p are real by (B1), so φ_p has exactly one zero in each interval $(k-1/2-r, k-1/2+r)$, $1 \leq k \leq n$, and no other zeros in $(0,n]$. Since $t_1 < t_2 < \cdots$ are the positive zeros in increasing order, $|t_k(p) - (k-1/2)| < r$ for $1 \leq k \leq n$ and all sufficiently large p. As r is arbitrary, $t_n(p) \to n - 1/2$. □

6. Uniqueness sets

In addition to (B1), (B2), (B3) and (B4), we use two classical facts about entire functions of exponential type.

- (P) (Plancherel and Pólya [11]) If g is an entire function of exponential type with $g \in L^p(\mathbb{R})$ for some $0 < p < \infty$, then g is bounded on $\mathbb{R}$.
- (C) (The Cartwright class [10, Lectures 16 and 17]) An entire function g of exponential type is in the Cartwright class if $\int_{\mathbb{R}} \log^+|g(t)|(1+t^2)^{-1}\,\mathrm{d}t < \infty$. Let $\sigma_\pm = h_g(\pm\pi/2)$, where h_g is the indicator of g. If

g is in the Cartwright class and not identically zero, then the zero-counting function $n(r)$ of g in $|z| < r$ satisfies $n(r)/r \to (\sigma_+ + \sigma_-)/\pi$, and

$$\log|g(z)| \le \sigma_+ \operatorname{Im} z + \frac{\operatorname{Im} z}{\pi} \int_{\mathbb{R}} \frac{\log|g(t)|}{|t-z|^2}\,\mathrm{d}t \tag{9}$$

for $\operatorname{Im} z > 0$, with the analogous inequality in the lower half-plane. If f and g are in the Cartwright class and f/g is entire, then f/g is in the Cartwright class and $h_{f/g} = h_f - h_g$.

The first lemma shows that a function in PW^p which vanishes on $\mathscr{Z}(\varphi_p) \cup \{0\}$ is a multiple of $x\varphi_p$ by a function of zero exponential type in each half-plane.

Lemma 7. *Suppose that $1 \le p < \infty$, let $g(z) = z\varphi_p(z)$, and suppose that f is a function in PW^p, $f \not\equiv 0$, which vanishes on $\mathscr{Z}(\varphi_p) \cup \{0\}$. Then $\omega = f/g$ is an entire function in the Cartwright class with $h_\omega(\pm\pi/2) \le 0$.*

Proof. The zeros of g are simple by (B1), so ω is entire. The functions f and g have exponential type and are bounded by a polynomial on $\mathbb{R}$, since f is bounded on $\mathbb{R}$ and $|\varphi_p| \le 1$. Hence both are in the Cartwright class, and $h_f(\pm\pi/2) \le \pi$. Since $t_n/n \to 1$ by (B1), the zero-counting function of g satisfies $n(r)/r \to 2$, so $h_g(\pi/2) + h_g(-\pi/2) = 2\pi$ by (C). The two values agree because g is real on $\mathbb{R}$, so $h_g(\pm\pi/2) = \pi$. By (C), ω is in the Cartwright class and $h_\omega(\pm\pi/2) = h_f(\pm\pi/2) - \pi \le 0$. □

For $p = 1$ we need the decay of φ_1 on $\mathbb{R}$ more precisely than (B2) gives it. We use the work of Bondarenko, Ortega-Cerdà, Radchenko and Seip [3] on the Hörmander–Bernhardsson function. Their function φ and constant $\mathscr{C}$ are defined by the extremal problem (2) with $p = 1$ [3, (1.1) and (1.2)], so $\varphi = \varphi_1$ and $\mathscr{C} = \mathscr{C}_1$ by uniqueness, and their τ_n is our $t_n(1)$. They factor $\varphi_1(z) = \Phi(z)\Phi(-z)$, where

$$\Phi(z) = \prod_{n=1}^{\infty} \left(1 + (-1)^n \frac{z}{t_n(1)}\right)$$

is an entire function with $\Phi(0) = 1$, and they prove [3, Theorem 1.1] that

$$\Phi(z) e^{1/(4\mathscr{C}_1 z)} = \frac{e^{-i\pi/4 + i\pi z/2}\,\Phi\big(1/(2\pi i \mathscr{C}_1 z)\big) + e^{i\pi/4 - i\pi z/2}\,\Phi\big(-1/(2\pi i \mathscr{C}_1 z)\big)}{2\sqrt{\pi \mathscr{C}_1}\, z} \tag{10}$$

for every $z \ne 0$. Their proof of (10) rests on the fact that $(n + 1/2 - t_n(1))_{n \ge 1}$ is in ℓ^2 [2, Theorem 1].

Lemma 8. *There is a constant M such that*

$$\left|\varphi_1(x) + \frac{\cos \pi x}{2\pi \mathscr{C}_1 x^2}\right| \le \frac{M}{|x|^3}$$

for every real x with $|x| \ge 1$. In particular $x\varphi_1(x)$ is not in $L^1(\mathbb{R})$.

Proof. Let $\kappa = 1/(2\sqrt{\pi\mathscr{C}_1})$, let x be real with $|x| \geq 1$, and put $w = 1/(2\pi\mathscr{C}_1 x)$. Since $\mathscr{C}_1 > 1/2$, we have $|w| < 1/\pi$. Let M_1 be the maximum of $|\Phi'|$ on the disc $|\zeta| \leq 1/\pi$. Since $\Phi(0) = 1$, we get $|\Phi(\pm iw) - 1| \leq M_1|w|$. Moreover $|e^{-1/(4\mathscr{C}_1 x)} - 1| \leq M_2/|x|$ for some constant M_2. Since $1/(2\pi i\mathscr{C}_1 x) = -iw$, (10) with $z = x$ gives

$$\begin{aligned} x\Phi(x) &= \kappa(e^{-i\pi/4 + i\pi x/2} + e^{i\pi/4 - i\pi x/2}) + O(|x|^{-1}) \\ &= 2\kappa \cos\frac{\pi(2x-1)}{4} + O(|x|^{-1}), \end{aligned}$$

where the implied constant does not depend on x. Replacing x by $-x$, we find that $-x\Phi(-x) = 2\kappa\cos(\pi(2x+1)/4) + O(|x|^{-1})$. The leading terms are bounded by 2κ, so multiplying the two expansions gives

$$\begin{aligned} -x^2\varphi_1(x) &= 4\kappa^2 \cos\frac{\pi(2x-1)}{4}\cos\frac{\pi(2x+1)}{4} + O(|x|^{-1}) \\ &= 2\kappa^2 \cos \pi x + O(|x|^{-1}), \end{aligned}$$

by the identity $2\cos a\cos b = \cos(a-b) + \cos(a+b)$ with $a - b = -\pi/2$ and $a + b = \pi x$. Since $2\kappa^2 = 1/(2\pi\mathscr{C}_1)$, this is the asserted estimate. Finally, for $x \geq 1$,

$$|x\varphi_1(x)| \geq \frac{|\cos \pi x|}{2\pi\mathscr{C}_1 x} - \frac{M}{x^2}.$$

The last term is integrable on $[1,\infty)$, while $\int_m^{m+1}|\cos \pi x|\,\mathrm{d}x = 2/\pi$ for every integer m, so $\int_1^\infty |\cos \pi x| x^{-1}\,\mathrm{d}x \geq \sum_{m\geq 1}(2/\pi)(m+1)^{-1} = \infty$. Hence $\int_{\mathbb{R}}|x\varphi_1(x)|\,\mathrm{d}x = \infty$. □

Proof of Theorem 3. Let $1 \leq p < \infty$, and suppose, to reach a contradiction, that f is a function in PW^p, $f \not\equiv 0$, which vanishes on $\mathscr{Z}(\varphi_p) \cup \{0\}$. Let $\omega = f/(x\varphi_p)$ be as in Lemma 7, and write $d(t) = \operatorname{dist}(t, \mathscr{Z}(\varphi_p))$.

Step 1: ω is a polynomial. Since f is bounded on $\mathbb{R}$, (B2) gives

$$\log|\omega(t)| \leq A_1 + \frac{4}{p}\log(1+|t|) + \log^+\frac{1}{|t|} + \log^+\frac{1}{d(t)}$$

for some constant A_1 and every real t outside $\mathscr{Z}(\varphi_p) \cup \{0\}$. Let $z = x + i$ with x real. By (9) and $h_\omega(\pi/2) \leq 0$, $\log|\omega(z)|$ is at most the Poisson integral $\pi^{-1}\int_{\mathbb{R}}\log|\omega(t)|\big((t-x)^2+1\big)^{-1}\,\mathrm{d}t$. We bound the contributions of the four terms. The constant contributes A_1. Since $\log(1+|t|) \leq \log(1+|x|) + \log(1+|t-x|)$, the second term contributes at most $(4/p)\log(1+|x|) + B_1$. The third term is integrable and the Poisson kernel is at most $1/\pi$, so it contributes at most an absolute constant. The fourth term vanishes unless $|t - \lambda| < 1$ for some $\lambda \in \mathscr{Z}(\varphi_p)$, and by the separation in (B1),

$$\frac{1}{\pi}\int_{\mathbb{R}}\frac{\log^+(1/d(t))}{(t-x)^2+1}\,\mathrm{d}t \leq \sum_{\lambda\in\mathscr{Z}(\varphi_p)}\frac{c_0}{1+(|x-\lambda|-1)_+^2} \leq B_2,$$

where $c_0 = \pi^{-1}\int_{-1}^1 \log^+(1/|s|)\,\mathrm{d}s$ and B_2 depends only on σ. Hence $|\omega(x+i)| \leq B(2+|x|)^{4/p}$, and likewise $|\omega(x-i)| \leq B(2+|x|)^{4/p}$, for every real x. Let

$k \geq 4/p$ be an integer and put $W(z) = \omega(z)(z+2i)^{-k}$. Then W is bounded on the boundary of the strip $|\operatorname{Im} z| \leq 1$ and of exponential type, so W is bounded in the strip by the Phragmén–Lindelöf principle, and in particular on $\mathbb{R}$. Since W is holomorphic in $\operatorname{Im} z > -2$, of exponential type, bounded on $\mathbb{R}$, and satisfies $h_W(\pi/2) \leq 0$, the Phragmén–Lindelöf principle in the upper half-plane shows that W is bounded in $\operatorname{Im} z \geq 0$. The same argument with $(z-2i)^{-k}$ gives a bound in $\operatorname{Im} z \leq 0$. Hence $|\omega(z)| \leq B'(2+|z|)^k$ in $\mathbb{C}$, and ω is a polynomial by Liouville's theorem.

Step 2: ω is constant. Suppose that ω has degree at least one. Then $|x\omega(x)| \geq c_1 x^2$ for $|x| \geq R$, with $c_1 > 0$. Since $p \geq 1$, we get $|f(x)|^p \geq c_1^p x^2 |\varphi_p(x)|^p$ for $|x| \geq \max(R, 1)$. As f is in $L^p(\mathbb{R})$, this contradicts (B3).

Step 3: the constants. By Steps 1 and 2, $f = cx\varphi_p$ for a constant $c \neq 0$, so $x\varphi_p$ is in $L^p(\mathbb{R})$. We show that this is false. For $p = 1$ this is Lemma 8. Suppose now that $p > 1$ and that $x\varphi_p$ is in $L^p(\mathbb{R})$. Then $\psi(z) = z\varphi_p(z)$ is in PW^p, and so is $\psi' = \varphi_p + z\varphi_p'$. Indeed, ψ' has exponential type at most π, and ψ is bounded on $\mathbb{R}$ by (P), so we may apply (4) to ψ and use Minkowski's inequality to get $\|\psi'\|_p \leq \pi \|\psi\|_p$. Hence $x\varphi_p' = \psi' - \varphi_p$ is in $L^p(\mathbb{R})$, while $|\varphi_p|^{p-1}$ is in $L^{p/(p-1)}(\mathbb{R})$. Since $\psi'(0) = \varphi_p(0) = 1$, (B4) applied to ψ' gives

$$\|\varphi_p\|_p^p = \int_{\mathbb{R}} \psi'(x)|\varphi_p(x)|^{p-2}\varphi_p(x)\,\mathrm{d}x = \|\varphi_p\|_p^p + \int_{\mathbb{R}} x\varphi_p'(x)|\varphi_p(x)|^{p-2}\varphi_p(x)\,\mathrm{d}x,$$

where the last integral converges absolutely by Hölder's inequality. Put $u = |\varphi_p|^p$. Since $p > 1$, the function $|\varphi_p|^{p-2}\varphi_p$, which we set equal to 0 at the zeros of φ_p, is continuous on $\mathbb{R}$. Hence u is continuously differentiable on $\mathbb{R}$ with $u' = p|\varphi_p|^{p-2}\varphi_p\varphi_p'$, so we have shown that $\int_{\mathbb{R}} xu'(x)\,\mathrm{d}x = 0$. On the other hand, u is even, so integration by parts gives

$$\int_{-R}^{R} xu'(x)\,\mathrm{d}x = 2Ru(R) - \int_{-R}^{R} u(x)\,\mathrm{d}x$$

for $R > 0$. Since $xu(x) \leq |x\varphi_p(x)|^p$ for $x \geq 1$, the function $xu(x)$ is integrable on $[1, \infty)$, so $\liminf_{R\to\infty} Ru(R) = 0$. Letting $R \to \infty$ along a suitable sequence, we find that $\int_{\mathbb{R}} xu'(x)\,\mathrm{d}x = -\|\varphi_p\|_p^p < 0$. Hence $x\varphi_p$ is not in $L^p(\mathbb{R})$, which is the desired contradiction. □

7. Concluding remarks

The size of A. The same certificate, with six patches plus the sinc patch $[1.9475, 2]$, proves Theorem 1 with $A = 3/50$; there the smallest margin is about $2 \cdot 10^{-4}$, ten times larger than for $A = 7/100$. With the test functions g_α, the method cannot reach $A = \gamma_E - 1/2 \approx 0.0772$, where γ_E is Euler's constant. Indeed, $J_q(2) > 1$ for $q < 2$, because $\varphi_2(z) = \operatorname{sinc} \pi z$ is the unique extremal function at $p = 2$, so the patch containing $p = 2$ must use g_1. Since $\log J_2$ is convex and vanishes at $p = 2$, condition (8) implies $A \leq 1/2 + D$ for every a, where

$$D = \frac{\mathrm{d}}{\mathrm{d}p} \log J_2(p)\Big|_{p=2} = \int_{-\infty}^{\infty} (\operatorname{sinc} \pi x)^2 \log|\operatorname{sinc} \pi x|\,\mathrm{d}x,$$

and $D = \gamma_E - 1$. To see this, substitute $t = \pi x$ and split $\log|\operatorname{sinc} t| = \log|\sin t| - \log|t|$. Since $\log|\sin t| \le 0$, Tonelli's theorem and $\sum_{k\in\mathbb{Z}}(t+k\pi)^{-2} = (\sin t)^{-2}$ give

$$\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\sin^2 t}{t^2}\log|\sin t|\,\mathrm{d}t = \frac{1}{\pi}\int_0^{\pi} \log\sin t\,\mathrm{d}t = -\log 2.$$

For the second part, let $F(\mu) = \int_0^\infty t^{\mu-1}\sin^2 t\,\mathrm{d}t$ for $-2 < \mu < 0$. Integrating by parts in $\int_0^\infty t^{\mu-1}(1-\cos 2t)\,\mathrm{d}t$ and using $\int_0^\infty t^{\nu-1}\sin t\,\mathrm{d}t = \Gamma(\nu)\sin(\pi\nu/2)$ for $-1 < \nu < 1$, we find that $F(\mu) = -2^{-\mu-1}\Gamma(\mu)\cos(\pi\mu/2)$. The expansions $\Gamma(-1+\varepsilon) = -1/\varepsilon + \gamma_E - 1 + O(\varepsilon)$ and $\cos(\pi(-1+\varepsilon)/2) = \pi\varepsilon/2 + O(\varepsilon^3)$ give $F(-1+\varepsilon) = \pi/2 - (\pi/2)(\gamma_E - 1 + \log 2)\varepsilon + O(\varepsilon^2)$, and differentiation under the integral sign shows that

$$\frac{2}{\pi}\int_0^{\infty} \frac{\sin^2 t}{t^2}\log t\,\mathrm{d}t = \frac{2}{\pi}F'(-1) = 1 - \gamma_E - \log 2.$$

Hence $D = -\log 2 - (1-\gamma_E-\log 2) = \gamma_E - 1$. In fact no constant $A > \gamma_E - 1/2$ is admissible in Theorem 1, whatever the test functions. Indeed, $\mathscr{C}_p \ge 1/J_2(p)$. In the other direction, suppose that $1 < p < \infty$ and let $p' = p/(p-1)$. The formula $f(0) = \int_{\mathbb{R}} f(x)\operatorname{sinc}\pi x\,\mathrm{d}x$ holds for f in PW^2, and hence for f in PW^p when $p \le 2$, since $PW^p \subset PW^2$ by (P). When $p > 2$, we apply it to $f((1-\varepsilon)z)\operatorname{sinc}(\pi\varepsilon z)$, which is in PW^2 and tends to f in $L^p(\mathbb{R})$ as $\varepsilon \to 0^+$, and use that $\operatorname{sinc}\pi x$ is in $L^{p'}(\mathbb{R})$. Together with Hölder's inequality, the formula gives $\mathscr{C}_p \le J_2(p')^{p-1}$. Both bounds equal 1 at $p = 2$, and the logarithms of both have derivative $-D$ there, since $\mathrm{d}p'/\mathrm{d}p = -1$ at $p = 2$. Hence $p \mapsto \log\mathscr{C}_p$ is differentiable at $p = 2$ with derivative $1 - \gamma_E$. The bound of Theorem 1 equals 1 at $p = 2$ and has logarithmic derivative $1/2 - A$ there, so it can hold for p slightly below 2 only if $1 - \gamma_E \le 1/2 - A$.

Integer samples of φ_p. The proof of Lemma 6 is quantitative. Applying Lemma 5 to φ_p itself, and using that every term satisfies $w_m\varphi_p(m) \ge -|w_m|$, we find that

$$|w_m|(1 - (-1)^m\varphi_p(m)) \le \pi^2 - c_p = O\Big(\frac{\log p}{p}\Big)$$

for every integer m. By Theorem 2 (a), φ_p is therefore close to $(-1)^m$ at each integer m, at a rate which is uniform for bounded m. Since $t_1(p) \ge 1/2$ for every p [4, Theorem 5.1], Theorem 2 shows that $t_1(p) \to 1/2$ from above; we do not know a rate for this convergence.

The case $p = 1$ of Theorem 3. (B2) and (B3) alone do not decide whether $x\varphi_1$ is integrable. The argument of Step 3 for $p > 1$ also applies when $p = 1$: the reproducing formula [4, Theorem 3.6] holds for $p = 1$ with $\operatorname{sgn}\varphi_1$ in place of $|\varphi_p|^{p-2}\varphi_p$, and $u = |\varphi_1|$ is absolutely continuous with $u' = \operatorname{sgn}(\varphi_1)\varphi_1'$ almost everywhere. We have nevertheless given the proof through Lemma 8, which rests on the description of φ_1 by Bondarenko, Ortega-Cerdà, Radchenko and Seip [2, 3], since the asymptotic formula is

of independent interest. The case $p = 1$ can also be obtained from their two-sided estimate [3, Lemma 2.5], which holds because $(n + 1/2 - t_n(1))$ is in ℓ^2 [2, Theorem 1]: it gives $|\varphi_1(x)| \geq c(1 + |x|)^{-2}$ whenever x has distance at least $1/4$ from $\mathscr{Z}(\varphi_1)$, and its second assertion states that every f in PW^1 which vanishes on $\mathscr{Z}(\varphi_1)$ is a constant multiple of φ_1. For general p, Step 1 of the proof of Theorem 3 shows that every f in PW^p, $1 \leq p < \infty$, which vanishes on $\mathscr{Z}(\varphi_p)$ has the form $P\varphi_p$ with P a polynomial of degree at most one. If P had degree one, then $|P(x)| \geq c|x|$ for large $|x|$ and some $c > 0$, so $x\varphi_p$ would be in $L^p(\mathbb{R})$, which Step 3 excludes. Hence P is constant for every $p \geq 1$.

On [4, Theorem 8.6]**.** The first version of this paper deduced Step 3 of the proof of Theorem 3 for $p > 1$ from [4, Theorem 8.6], which states that $\sum_{n\geq 1} n^{-2}|\varphi_p(\mu_n)|^{-p} < \infty$ for $1 < p < \infty$, where $\mu_n = (t_n + t_{n+1})/2$. As printed, this fails for $p = 2$: then $t_n = n$ and $|\varphi_2(\mu_n)|^2 = \pi^{-2}(n + 1/2)^{-2}$, so the terms of the series tend to π^2. The theorem is deduced in [4] from the summability of $\|e_n\|^{-2}t_n^{-2}$, where $e_n(z) = z\varphi_p(z)/(z - t_n)$ and the norm is that of $L^2(|\varphi_p|^{p-2}\,\mathrm{d}x)$, together with a comparison of $\|e_n\|^2$ with $|\varphi_p(\mu_n)|^p$. For $p = 2$ one has $\|e_n\|^2 = 1$ for every n, which is comparable to $t_n^2|\varphi_2(\mu_n)|^2$ rather than to $|\varphi_2(\mu_n)|^2$. If $\|e_n\|^2$ is compared with $t_n^2|\varphi_p(\mu_n)|^p$ instead, the argument leads to the series with n^{-4} in place of n^{-2}. That version would give Step 3 only for $p \geq 2$, where it already follows from (B3), so the present proof of Step 3 does not use it.

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